

Parallel Quantum Dynamics

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**Self-centric parallelization (easy spatial decomposition)
of a partial-differential-equation solver
as a ‘boundary condition’**



Self-Centric (SC) Parallelization

- SC is the easiest serial-to-parallel migration path *via* single-program multiple-data (SPMD) programming
 1. Take a serial code
 2. Each MPI rank only works on a spatial subsystem
 3. Boundary information obtained from neighbor ranks
 4. Long-range interaction by real-space multigrids; scalability behavior is similar to short-ranged (see slides 7-8 in <https://aiichironakano.github.io/cs653/02-04DC-slide.pdf>)

T. Razakh et al., [*Gordon Bell Prize finalist*](#) ('25)
S. C. Tiwari et al., [*ACM HPCAsia2020, Best Paper*](#) ('20)
F. Shimojo et al., [*J. Chem. Phys.* **140**, 18A529](#) ('14)
K. Nomura et al., [*IEEE/ACM Supercomputing, SC14*](#) ('14)
A. Nakano, [*Comput. Phys. Commun.* **83**, 181](#) ('94)



Quantum Dynamics Program: qd1.c

```

for step = 1 to NSTEP
  pot_prop():  $\psi_j \leftarrow \exp(-iV_j\Delta t/2)\psi_j$  ( $j \in [1, NX]$ )
  kin_prop( $\Delta t/2$ )
  kin_prop( $\Delta t$ )
  kin_prop( $\Delta t/2$ )
  pot_prop():  $\psi_j \leftarrow \exp(-iV_j\Delta t/2)\psi_j$  ( $j \in [1, NX]$ )
  
```

*cf. velocity-Verlet
half-time acceleration*

$$\begin{aligned}
 \psi(t + \Delta t) &\leftarrow \exp(-iV\Delta t/2) \exp(-iT_x\Delta t) \exp(-iV\Delta t/2) \psi(t) \\
 &= e^{-iV\Delta t/2} U_x^{(\text{half})} U_x^{(\text{full})} U_x^{(\text{half})} e^{-iV\Delta t/2} \psi(t)
 \end{aligned}$$

```

kin_prop( $\Delta$ )
periodic_bc():  $\psi_0 \leftarrow \psi_{NX}$ ;  $\psi_{NX+1} \leftarrow \psi_1$ 
for  $\forall j \in [1, NX]$ 
   $\psi_j \leftarrow \text{blx}(\Delta)_j \psi_{j-1} + \text{al}(\Delta)_j \psi_j + \text{bux}(\Delta)_j \psi_{j+1}$ 
  
```

$$\begin{cases} \varepsilon_n^+ = \frac{1}{2} \left[\exp\left(-\frac{i\Delta t}{n}(a+b)\right) + \exp\left(-\frac{i\Delta t}{n}(a-b)\right) \right] \\ \varepsilon_n^- = \frac{1}{2} \left[\exp\left(-\frac{i\Delta t}{n}(a+b)\right) - \exp\left(-\frac{i\Delta t}{n}(a-b)\right) \right] \end{cases}$$

$$\exp(-i\Delta t T_x) \cong U_x^{(\text{half})} U_x^{(\text{full})} U_x^{(\text{half})} = \begin{bmatrix} \varepsilon_2^+ & \varepsilon_2^- & & & \\ \varepsilon_2^- & \varepsilon_2^+ & & & \\ & & \varepsilon_2^+ & \varepsilon_2^- & \\ & & \varepsilon_2^- & \varepsilon_2^+ & \\ & & & \ddots & \\ & & & & \varepsilon_2^+ & \varepsilon_2^- \\ & & & & \varepsilon_2^- & \varepsilon_2^+ \end{bmatrix} \begin{bmatrix} \varepsilon_1^+ & & & & \\ \varepsilon_1^+ & \varepsilon_1^- & & & \\ \varepsilon_1^- & \varepsilon_1^+ & & & \\ & & \ddots & & \\ & & & \varepsilon_1^+ & \varepsilon_1^- \\ & & & \varepsilon_1^- & \varepsilon_1^+ \end{bmatrix} \begin{bmatrix} \varepsilon_2^+ & \varepsilon_2^- & & & \\ \varepsilon_2^- & \varepsilon_2^+ & & & \\ & & \varepsilon_2^+ & \varepsilon_2^- & \\ & & \varepsilon_2^- & \varepsilon_2^+ & \\ & & & \ddots & \\ & & & & \varepsilon_2^+ & \varepsilon_2^- \\ & & & & \varepsilon_2^- & \varepsilon_2^+ \end{bmatrix}$$

<https://aiichironakano.github.io/phys516/03QD-slide.pdf>

Quantum Dynamics Computation

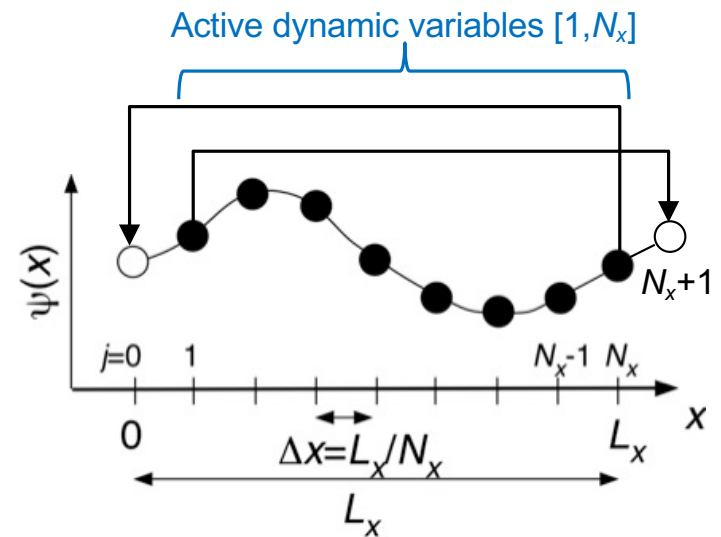
- **Essence:** Keep updating wave-function values mesh point-by-point as a function of those on the nearest-neighbor mesh points

$$\psi_j(t+1) \leftarrow \mathcal{F}(\psi_{j-1}(t), \psi_j(t), \psi_{j+1}(t)) \quad (j \in [1, NX])$$

- **Periodic boundary condition *via* augmentation**

```
void periodic_bc() {  
    int s;  
    for (s=0; s<=1; s++) {  
        psi[0][s] = psi[NX][s];  
        psi[NX+1][s] = psi[1][s];  
    }  
}
```

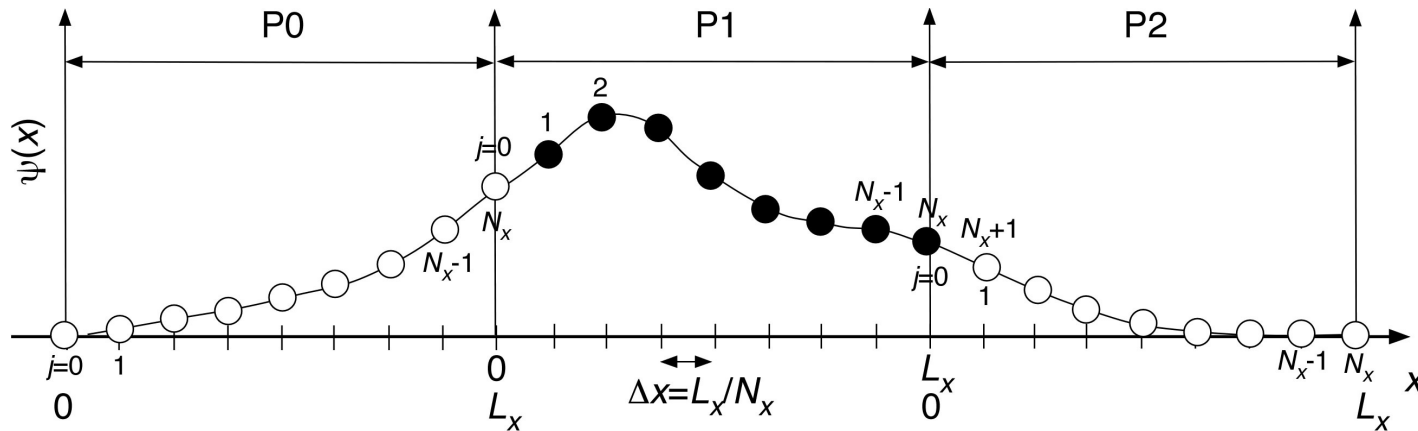
- **Often sufficient just to understand computational characteristics for parallelizing a serial code**



<https://aiichironakano.github.io/cs596/src/qd>

SC Parallelization

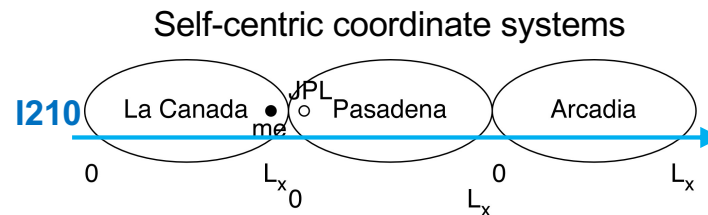
- Self-centric spatial decomposition



- Local & global coordinates

$$\begin{cases} x_j = j\Delta x \\ x_j^{(\text{global})} = j\Delta x + pL_x \end{cases}$$

offset



- Global coordinates only in `init_prop()` & `init_wavfn()`
initialization of simulation

Boundary Wave Function Caching

- Parallelized periodic_bc()**

```

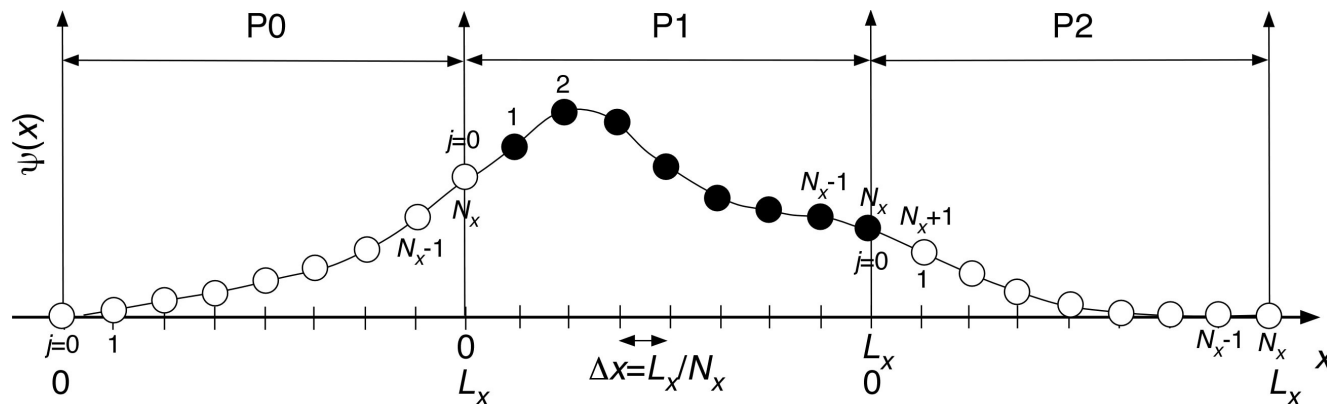
MPI_Comm_rank(MPI_COMM_WORLD, &myid);
MPI_Comm_size(MPI_COMM_WORLD, &nproc);

plw = (myid-1+nproc)%nproc; // Lower partner process
pup = (myid+1)%nproc; // Upper partner process

/* Cache boundary wave function value at the lower end */
dbuf[0:1] ← psi[NX][0:1]; I. Message (1D array) composition
Send dbuf to pup; II. Message passing
Receive dbufr from plw; III. Message storing
psi[0][0:1] ← dbufr[0:1];

/* Cache boundary wave function value at the upper end */
dbuf[0:1] ← psi[1][0:1];
Send dbuf to plw;
Receive dbufr from pup;
psi[NX+1][0:1] ← dbufr[0:1];

```



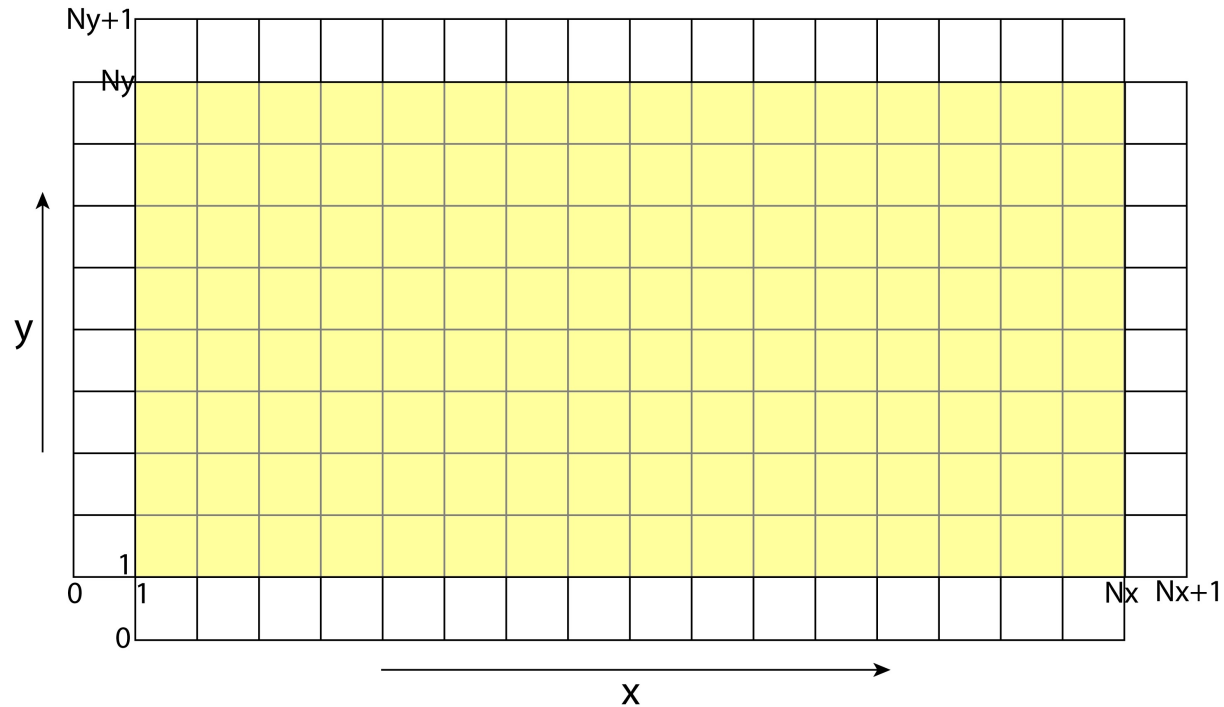
Multidimensional Parallelization

- Parallelized `periodic_bc()`

for $\forall$ directions

send front row `psi(...,1 or  $N_\alpha$ ,...)` to forward neighbor

receive back appendage `psi(..., $N_\alpha+1$  or 0,...)` from back neighbor



Multidimensional Parallelization

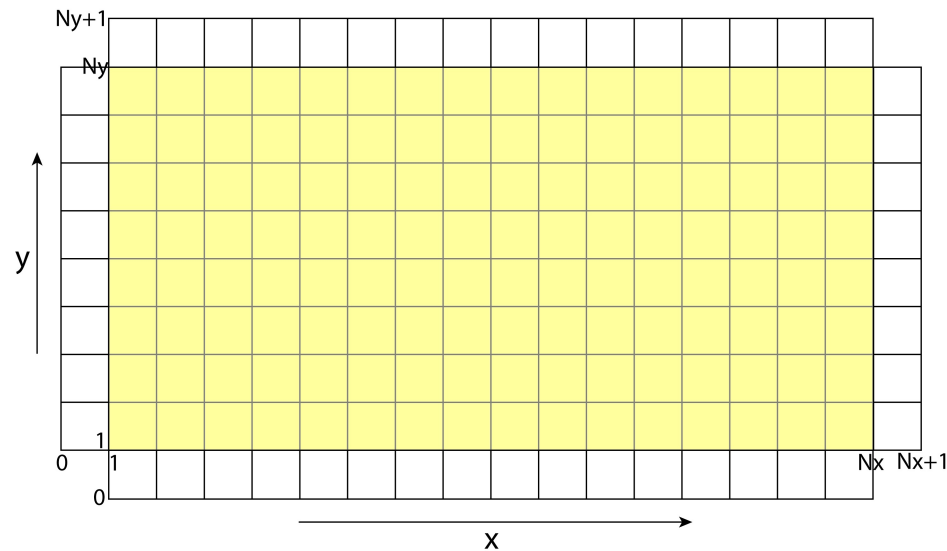
- Message composition

$dbuf \leftarrow psi(i_b : i_e, j_b : j_e, k_b : k_e)$
 $psi(i'_b : i'_e, j'_b : j'_e, k'_b : k'_e) \leftarrow dbufr$

(Example) x-low direction

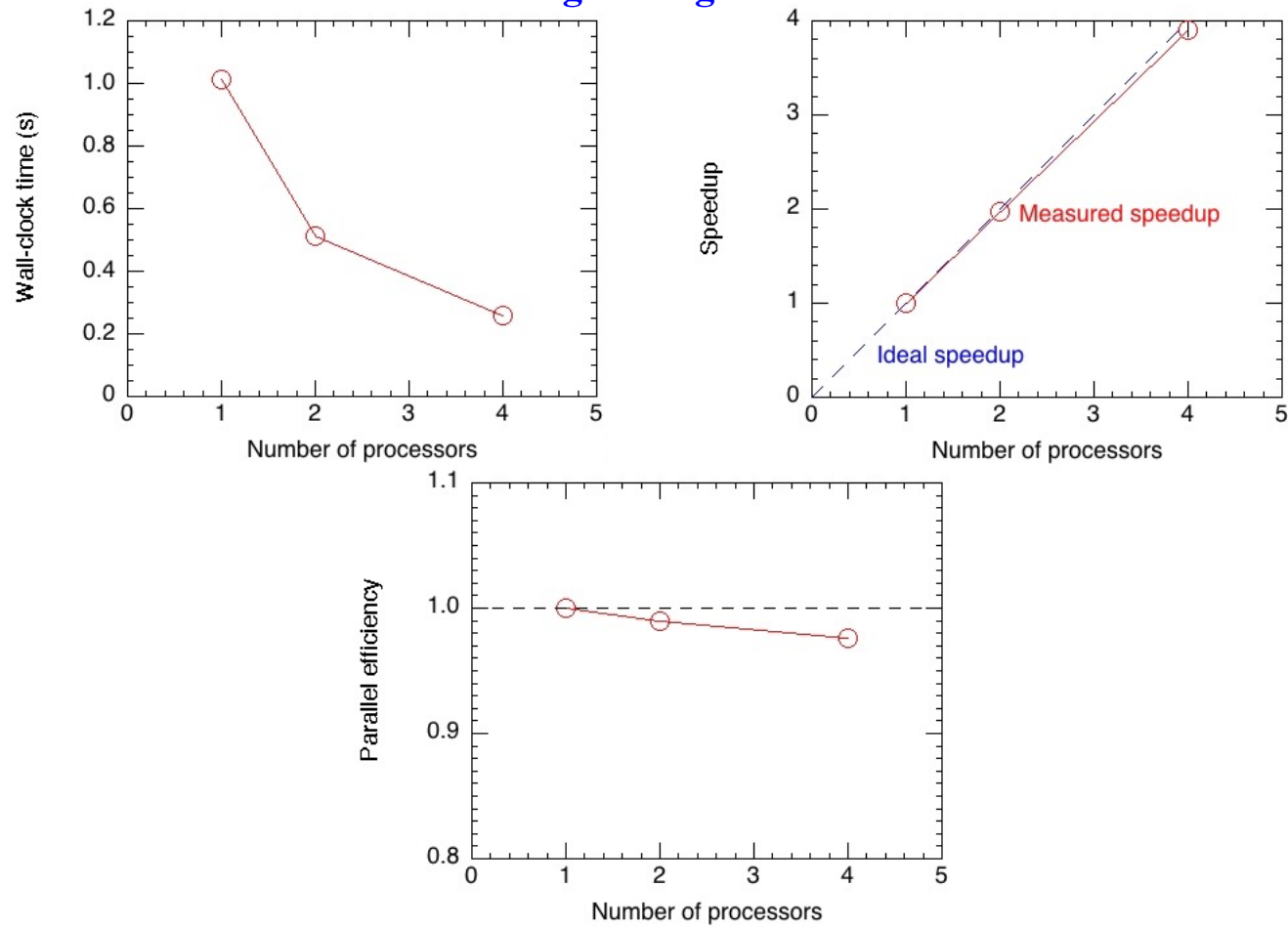
$i_b = 1, i_e = 1, j_b = 1, j_e = N_y, k_b = 1, k_e = N_z$

$i'_b = N_x + 1, i'_e = N_x + 1, j'_b = 1, j'_e = N_y, k'_b = 1, k'_e = N_z$

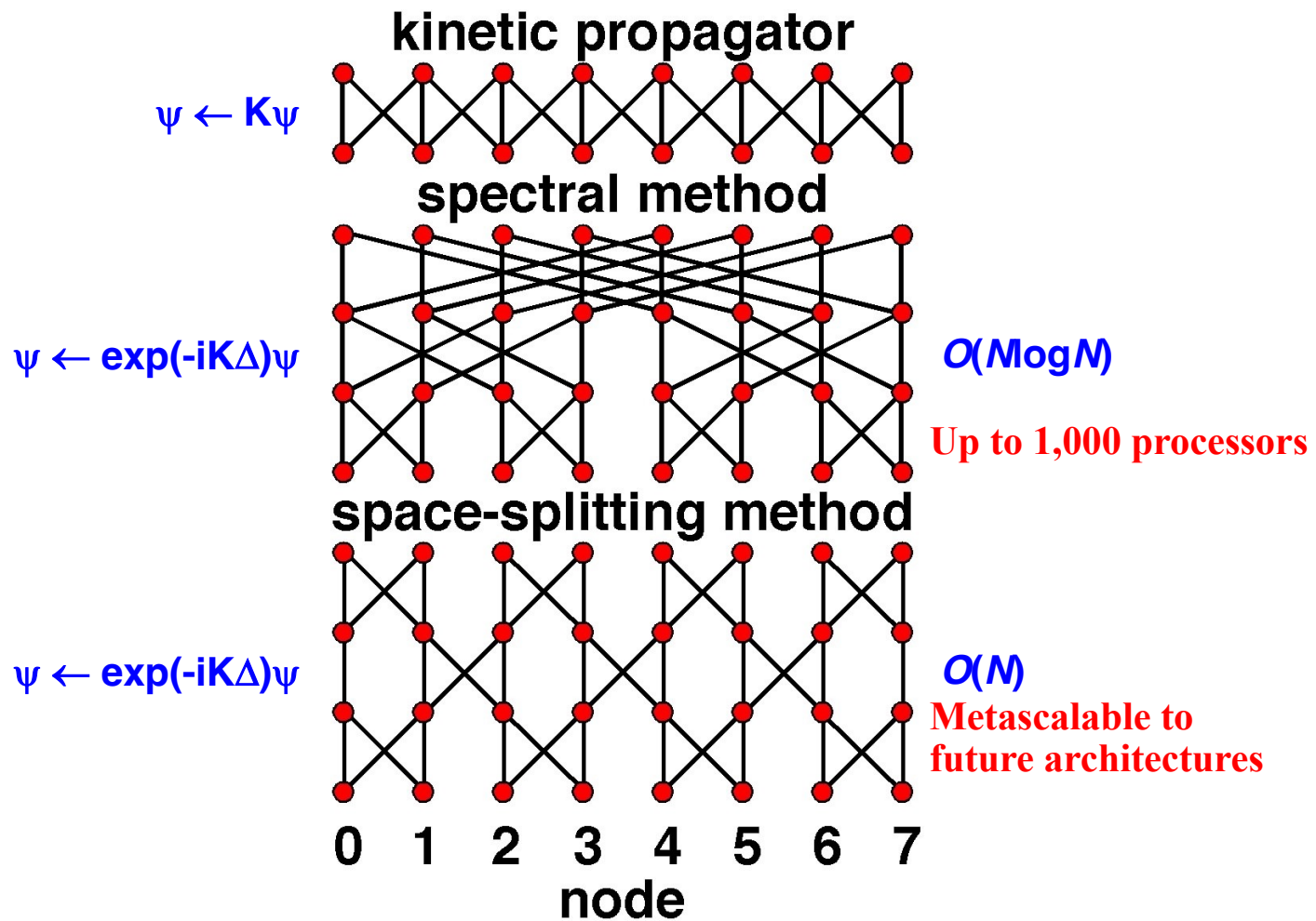


Parallel QD Results

Strong scaling results



Parallel QD Communications



Parallel QD Algorithms

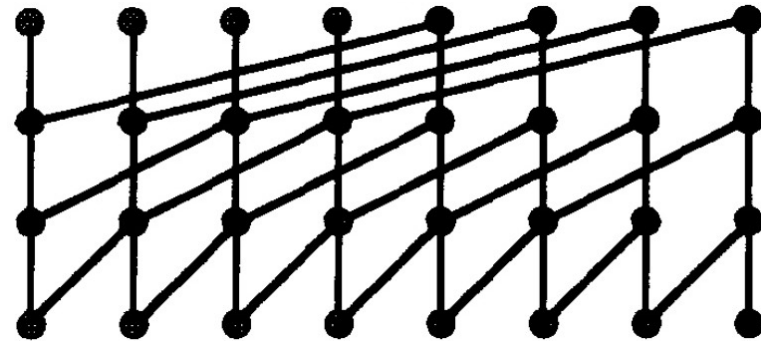
- Not all algorithms are scalable on parallel computers
- Implicit solvers (*e.g.*, Crank-Nicholson method) are numerically stable but less scalable due to sequential dependence

$$\psi(t + \Delta t) \leftarrow \exp\left(-\frac{i}{\hbar} \hat{H} \Delta t\right) \psi(t) \cong \frac{1 - \frac{i}{2\hbar} \hat{H} \Delta t}{1 + \frac{i}{2\hbar} \hat{H} \Delta t} \psi(t) + O((\Delta t)^3)$$

$$\overbrace{\left(1 + \frac{i}{2\hbar} \hat{H} \Delta t\right)}^A \overbrace{\psi(t + \Delta t)}^x = \overbrace{\left(1 - \frac{i}{2\hbar} \hat{H} \Delta t\right)}^b \psi(t)$$

$$\alpha x_{i-1} + \beta x_i + \alpha x_{i+1} = b_i$$

$$\Rightarrow x_{i+1} \leftarrow \frac{1}{\alpha} b_i - \frac{\beta}{\alpha} x_i - x_{i-1}$$



- Sequential recursion needs be converted to divide-&-conquer (recursive doubling) for parallelization